

Wei-Kai Li Academic CV

Professor / Doctoral Supervisor / Taishan Scholars Young Expert

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School of Computer Science and Artificial Intelligence, Shandong Jianzhu University / School of Mathematics and Statistics, Chongqing Jiaotong University / Panoramic Medical Imaging Center



Education

1. 2018–2022 / PhD / Computer Science and Technology / Nanjing University of Aeronautics and Astronautics
2. 2015–2018 / Master's / Computer Technology and Applications / Chongqing Jiaotong University
3. 2011–2015 / Bachelor's / Information and Computing Science / Chongqing Jiaotong University

Research Projects

1. Unsupervised Domain Adaptation beyond Domain Alignment and Its Extensions / NSFC Young Scientists Fund / Principal Investigator
2. Advanced School in Applied Machine Learning / NSFC International Cooperation and Exchange / Principal Investigator
3. Multimodal Neurophysiological Mechanisms of Heterogeneous Brain Networks in Depression and Digital-Twin Closed-Loop TMS Robotics / Shandong Major Basic Research Project / Principal Investigator
4. Exploration of Unsupervised Domain Adaptation beyond Domain Alignment / MIIT Key Laboratory Open Project / Principal Investigator
5. Early Identification of Parkinson's Disease Using PET/MR Multimodal Brain Networks / Hubei Key Laboratory of Molecular Imaging / Principal Investigator
6. Multiscale Analysis of Autism Using Multimodal Imaging / Chongqing Education Commission Young Research Fund / Principal Investigator
7. Brain Network Disease Diagnosis Based on Multicenter Transfer Learning / Chongqing Natural Science Foundation General Project / Principal Investigator
8. Probabilistic Synchronization and Secure Control of Stochastic Complex Networks / Chongqing Natural Science Foundation Joint Fund / Researcher
9. Investigating and Extending the Effectiveness of Self-Supervised Learning / NSFC General Program / Researcher

Publications

- [1] QIN H, HONG G, MA D, et al. Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View[J]. CNS Neuroscience & Therapeutics, 2026, 32(9): e71158.
- [2] ZHENG Z, CHEN X, JIANG H, et al. Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI[PP/OL]. arXiv (2026-07)[2026-10-05]. <https://arxiv.org/abs/2607.29531>.
- [3] YANG J, LI W, QIAN K, et al. Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction[J]. Neurocomputing, 2026: 134626.
- [4] CUI L, WU Y, TANG Y, et al. Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation[J]. IEEE Transactions on Computational Social Systems, 2026.
- [5] MA D, PENG L, ZHANG L, et al. Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis[J]. Frontiers in Psychiatry, 2026, 17: 1846996.
- [6] WANG H, LENG J, LI S, et al. TumorAL: Evidence-aware active learning for 3D tumor segmentation[J]. Neurocomputing, 2026: 134239.

- [7] RUAN Y, GUAN L, PENG L, et al. Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns[J]. *CNS Neuroscience & Therapeutics*, 2026, 32(6): e70951.
- [8] WANG Q, LI W, WAN J, et al. Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease[J]. *Journal of Cardiovascular Development and Disease*, 2026, 13(6): 228.
- [9] CHEN X, ZOU Q, CAI Q, et al. Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq[J]. *PLOS Computational Biology*, 2026, 22(5): e1014223.
- [10] YANG J, ZHANG J, QIAN K, et al. Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework[J]. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, 40(32): 27529.
- [11] WU Y, LI Y, WEI H, et al. Test-time few-shot object detection via dynamic prototype fusion[J]. *IEEE Transactions on Cybernetics*, 2026.
- [12] WANG Q, XIA Y, WEI L, et al. Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect[J]. *Cardiology in the Young*, 2026: 1-11.
- [13] MA H, GAN HS, MI H, et al. Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation[C]//2025 IEEE International Conference on Big Data (BigData). 2025: 7503-7510.
- [14] LI D, YAO J, LI W, et al. More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease[J]. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 2025: 111578.
- [15] SONG H, QIN H, LI W. Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis[C]//2025 International Conference on Cyber-Physical Social Intelligence (CPSI). 2025: 1-6.
- [16] LU Y, TIAN N, LI W. MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning[C]//2025 International Conference on Cyber-Physical Social Intelligence (CPSI). 2025: 1-6.
- [17] RUAN Y, MA H, MA D, et al. Low-light image enhancement using dual cross attention[J]. *Engineering Applications of Artificial Intelligence*, 2025, 159: 111501.
- [18] MOU X, LI W, LI B, et al. Auto-weighted projective one-step multi-view clustering[J]. *Neurocomputing*, 2025: 131960.
- [19] LI W, TIAN N, LI Y, et al. Revisiting Multi-Modal Alignment: In Distribution View[C]//2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). 2025: 6325-6330.
- [20] XU X, LI Y, LAN H, et al. Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis[J]. *BMC psychiatry*, 2025, 25(1): 826.
- [21] YANG Y, LI B, CAO W, et al. Cross-modal medical image generation from MRI to PET using robust generative adversarial network[J]. *Expert Systems with Applications*, 2025: 129287.
- [22] CUI L, WU Y, YING T, et al. Leveraging multi-source and heterogeneous signals for fatigue detection[PP/OL]. *arXiv (2025-07)[2026-10-05]*. <https://arxiv.org/abs/2507.16859>.
- [23] MA Q, CHEN B, SEIFERT R, et al. PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended pelvic lymph node dissection by preoperative nomograms[J]. *European Journal of Nuclear Medicine and Molecular Imaging*, 2025, 52(6): 2063-2075.
- [24] CHEN X, LIN S, CHEN X, et al. Timestamp calibration for time-series single cell rna-seq expression data[J]. *Journal of Molecular Biology*, 2025, 437(9): 169021.
- [25] CHEN X, LIN D, LI Z, et al. Iterative neural networks for improving memory capacity[J]. *Neural Networks*, 2025, 182: 106936.
- [26] SUN D, LI W, YU H, et al. The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods[J]. *Journal of Cyber-Physical-Social Intelligence*, 2024, 3(1): 10-10.
- [27] CHEN S, CHEN X, LI W. An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [28] XIAO X, CHEN X, LI W. Improved domain-adversarial neural networks with mixup and pseudo-labeling[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [29] JIANG Y, LI W, CHEN X. Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

- [30] YANG Y, LI B, LI W, et al. Synthesizing A β -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer's disease[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [31] QIN H, XU X, LI W. Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [32] SUN D, CHEN X, LI W. Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [33] TIAN N, XU X, LI W. Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [34] DENG J, GUAN L, LI W. Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.
- [35] MOU L, XIAO X, CAO W, et al. Efficient and accurate capsule networks with b-spline-based activation functions[C]//2024 International Conference on New Trends in Computational Intelligence. 2024.
- [36] CHEN H, XU J, LI W, et al. The characteristic patterns of individual brain susceptibility networks underlie Alzheimer's disease and white matter hyperintensity-related cognitive impairment[J]. Translational Psychiatry, 2024, 14(1): 177.
- [37] LI W, WEI H, WU Y, et al. TIDE: Test-time few-shot object detection[J]. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024, 54(11): 6500-6509.
- [38] ZHOU W, SUN M, XU X, et al. Multipattern graph convolutional network-based autism spectrum disorder identification[J]. Cerebral Cortex, 2024, 34(3): bhac064.
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- [43] YANG J, XU X, SUN M, et al. Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data[J]. Cerebral Cortex, 2024, 34(1): bhac477.
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- [46] DING Y, XU X, PENG L, et al. Wavelet transform-based frequency self-adaptive model for functional brain network[J]. Cerebral Cortex, 2023, 33(22): 11181-11194.
- [47] ZHENG H, HU J, LI W. PageRank algorithm based on dynamic damping factor[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.
- [48] LI T, CHEN X, LI W. An improved AdaBoost method in imbalanced data Learning[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.
- [49] CHENG Z, CHEN X, LI H, et al. Minimum error entropy high-order extended Kalman filter[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.
- [50] LI W, AL-AMIN MM, NAIR A. New insights into the disorder of brain connectivity in schizophrenia[J]. Frontiers in Neuroimaging, 2023, 2: 1266695.
- [51] LI W, TANG Y, PENG L, et al. The reconfiguration pattern of individual brain metabolic connectome for Parkinson's disease identification[J]. MedComm, 2023, 4(4): e305.
- [52] WANG Y, CHEN Q, LIU Y, et al. TIToK: A solution for bi-imbalanced unsupervised domain adaptation[J]. Neural Networks, 2023, 164: 81-90.

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- [76] ZHOU Y, QIAO L, LI W, et al. Simultaneous estimation of low-and high-order functional connectivity for identifying mild cognitive impairment[J]. Frontiers in neuroinformatics, 2018, 12: 337293.
- [77] LI W, WANG Z, ZHANG L, et al. Remodeling Pearson's correlation for functional brain network estimation and autism spectrum disorder identification[J]. Frontiers in neuroinformatics, 2017, 11: 55.
- [78] YU J, WANG X, TU S, et al. Kola: Carefully benchmarking world knowledge of large language models[PP/OL]. arXiv (2023-06)[2026-10-05]. <https://arxiv.org/abs/2306.09296>.
- [79] CHEN S, CHEN X, CAO W, et al. An Improved Efficient Algorithm for Spiking Neural Network Through Dataset Distillation[PP/OL]. SSRN[2026-10-05]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4836784.
- [80] XIAO X, CHEN X, LI W, et al. Graph Adversarial Discriminative Domain Adaptation with Category-Level Alignment[PP/OL]. SSRN[2026-10-05]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4898772.

Academic Service

- 1. Program Chair, CPSI 2027
- 2. Co-Program Chair, CPSI 2026
- 3. Area Chair, IEEE SMC 2026
- 4. Area Chair, IFAC CPHS 2020
- 5. Area Chair, ICCSI 2022–2025
- 6. Convener, Embodied Intelligence Conference Parallel Forum
- 7. Youth Editorial Board, Journal of Data Acquisition and Processing
- 8. Youth Editorial Board, Meta-Radiology
- 9. Youth Editorial Board, Command, Control and Simulation
- 10. Guest Editor, Frontiers in Neuroscience
- 11. Guest Editor, Frontiers in Aging Neuroscience
- 12. Guest Editor, Frontiers in Cell and Developmental Biology
- 13. Guest Editor, Frontiers in Pharmacology
- 14. Outstanding Reviewer, Journal of Data Acquisition and Processing
- 15. Youth Editorial Board, Aging & Disease
- 16. Youth Editorial Board, Brain-X
- 17. Youth Editorial Board, Cog
- 18. Youth Editorial Board, Artificial Intelligence Science and Engineering
- 19. Corresponding Member, CAAI Machine Learning Committee
- 20. Deputy Secretary-General, CICC Embodied Intelligence Committee
- 21. Member, CAA Human–Machine Education Committee
- 22. Member, CSS Virtual Technology and Applications Committee
- 23. Committee Member, Youth Workshop on Medical Imaging
- 24. Council Member, Chongqing Mathematical Society

Personal Recognition

- 1. 2026 / IEEE SMC 2026 Best Conference Paper Award Finalist
- 2. 2026 / CPSI Best Paper Award
- 3. 2025 / Taishan Scholars Young Expert

4. 2025 / CPSI Best Paper Award
5. 2025 / Outstanding Faculty Advisor, China Robot and AI Competition
6. 2023 / ICCSI Best Paper Finalist Award
7. 2022 / National Scholarship for Doctoral Students

Student Awards

1. 2026 / Jingxuan Cui and Coauthors: CPSI 2026 Best Paper Award
2. 2025 / First Prize, China Robot and AI Competition National Finals
3. 2025 / First Prize, China Robot and AI Competition Chongqing Selection
4. 2025 / National Third Prize, National Digital Art Design Competition
5. 2025 / MCM/ICM Honorable Mention and Meritorious Winner
6. 2025 / First Prize, China International College Students' Innovation Competition: Paying Xianzhi
7. 2025 / First Prize, China International College Students' Innovation Competition: Yingluo Zhizhen
8. 2025 / Second Prize, iCAN Campus Competition
9. 2025 / Third Prize, China Graduate Electronics Design Competition, Southwest Region
10. 2024 / MCM/ICM Honorable Mention and Meritorious Winner
11. 2024 / First and Third Prizes, China International College Students' Innovation Competition, Campus Round
12. 2024 / Second Prize, China Robot and AI Competition National Finals
13. 2024 / Second Prize, Chongqing Robot Competition
14. 2024 / First Prize, National Business Elite Challenge Finals
15. 2024 / Third Prize, National Intelligent Transportation Innovation and Entrepreneurship Competition
16. 2024 / Third Prize, China Graduate Electronics Design Competition
17. 2024 / Second Prize, University Entrepreneurship Competition
18. 2024 / First Prize, RAICOM Robot Developer Competition
19. 2023 / First and Second Prizes, National Undergraduate Mathematical Contest in Modeling, Chongqing
20. 2023 / Third Prize, China Internet+ College Students' Innovation and Entrepreneurship Competition, Campus Round
21. 2023 / Third Prize, China Software Cup Western Division